

Mohammadreza Nemati

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EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Fine-tuned multimodal foundation models using SFT, DPO, and GRPO for planner, reasoner, grounder, verifier, and error-recovery components in real-device automotive HMI software testing.
- Compressed VLMs using pruning, knowledge distillation, and quantization-aware training for constrained deployment, improving inference efficiency for agentic AI workloads.
- Built multimodal GUI-agent system for real-device Android HMI software testing, combining screen understanding, element grounding, tool-use actions, action verification, and error recovery across in-vehicle app workflows.
- Developed long-horizon planning workflows for automotive HMI test agents, including dynamic task decomposition, app navigation, action execution, and completion-evidence tracking for multi-step validation goals.
- Served custom VLM checkpoints with vLLM continuous batching and optimized attention backends to improve throughput and latency for agentic AI inference workloads.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built VLM-based visual grounding system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language object grounding and segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

Agentic AI: GUI Agents, Long-Horizon Planning, Task Decomposition, Action Verification, Error Recovery, Multi-Agent Workflows

LLM / VLM Systems: Python, PyTorch, Vision-Language Models, Multimodal Reasoning, Visual Grounding, Prompting, Model Evaluation

Evaluation / Optimization: Trace Logging, Agent Evaluation, Latency/Throughput Tuning, Quantization, Knowledge Distillation, Pruning

Deployment / Frameworks: vLLM, CUDA, Model Serving, Distributed Training (FSDP/DDP), LangGraph, Agentic RAG

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017